

HEAVY-LIFT UAV MAIN POWER PATH (48V HYBRID DC BUS)

1. HYBRID GENERATOR & RECTIFIER SYSTEM
 - Output: 3.6 kW Continuous (48V DC, 75A Max)
 - Fuel: Gasoline 2-Stroke Gen
 - Cable Rating: 48V_BUS, MAX 75A

2. BUFFER BATTERY & BMS FAILSAFE
 - Battery: 12S LiPo (44.4V – 50.4V, 4500mAh 45C)
 - Failsafe BMS Parallel Coupling (Ideal Diode)
 - Cable Rating: BAT_BUS, MAX 100A PEAK

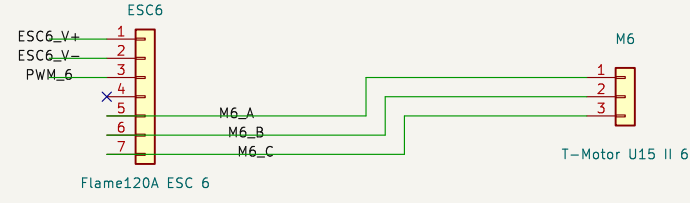
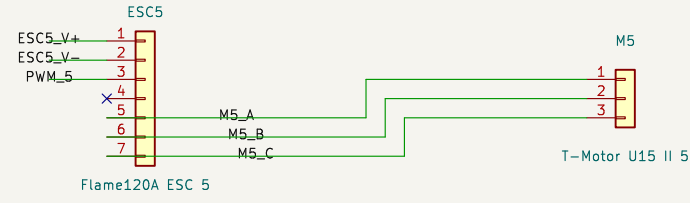
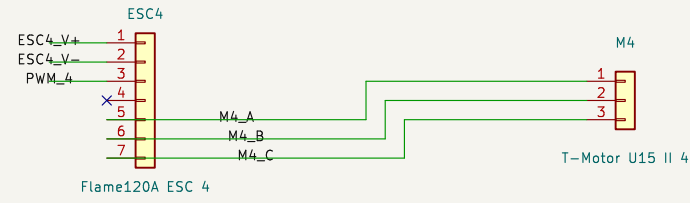
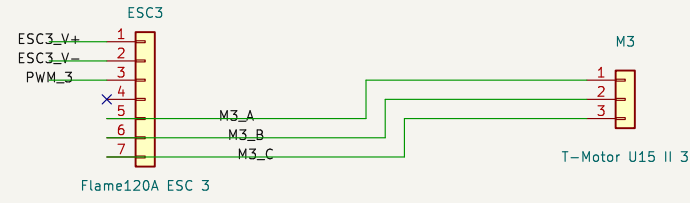
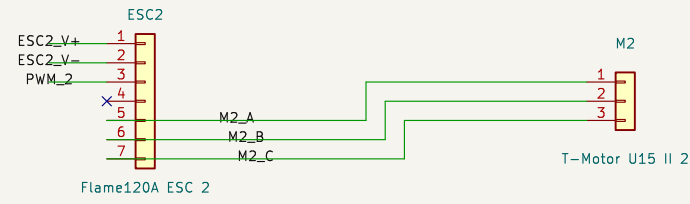
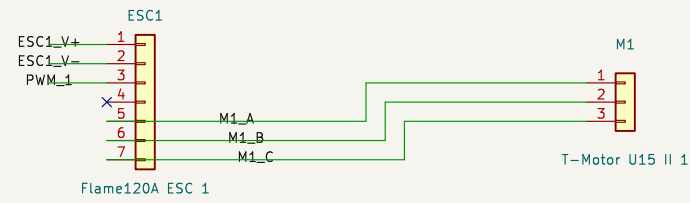
3. MAIN POWER DISTRIBUTION BOARD (PDB)
 - Input: 48V DC Bus (Total Hover Power: 3,460.1 W @ 72.1A Total)
 - 6x ESC Branch Outputs: 48V DC, 12.02A Hover / 60A Peak per Branch

3. REGULATED RAIL OUTPUTS
 - REG1: 12V 10A Rail for Video Tx / Camera Gimbal
 - REG2: 5V 10A Rail for Flight Controller & Jetson Orin

AVIONICS & CONTROL SIGNAL ARCHITECTURE

2. COMPANION COMPUTER
 - Model: Nvidia Jetson Orin Nano
 - High-Level autonomy and payload interface
 - Powered by DC-DC 5V Regulator

1. FLIGHT CONTROLLER (Pixhawk 6X)
 - PWM Outputs [PWM_1 .. PWM_6] -> 6x ESC Signal Inputs
 - Telemetry 1 -> 915 MHz RF Telemetry Radio (1W Output)
 - GPS / RTK -> Dual Neo-3 GPS/RTK Receivers (Serial GPS1 + CAN GPS2)
 - CAN Bus [CAN1_H/L] -> Hybrid Generator ECU Telemetry, BMS, and GPS2
 - Companion Computer -> Ethernet MAVLink to Nvidia Jetson Orin Nano



Safety Factor & Peak Rating: 120A Flame ESCs, 100A BMS Battery Failsafe Coupling		
Power Distribution: 48V Bus, 72.1A Total Hover Current (12.02A/Branch), 75A Gen Max Output		
Locked Baseline (DESIGN_LOCK.md): 6 Arms @ 60 deg (L=1.12m), 40in Props, 3460.1W Hover Power,		
Flight Systems Division		
Sheet: /		
File: power_and_signal_schematic.kicad_sch		
Title: Heavy-Lift Gas-Electric Hybrid Hexacopter – Power & Signal Schematic		
Size: A3	Date: 2026-07-22	Rev: 1.8
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